PAPER_587 — Inflationary Epoch Details from UQFF Grinding

CP4 Class: #174 `UQFFInflationaryEpochDetailsCalculator` **Session:** 157 **Cross-refs:** PAPER_586 (Big Bang), PAPER_589 (Dark Energy), PAPER_583 (6-Form) **Source:** grok_share_4cef778c78b8.txt

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Abstract

This paper presents a UQFF analysis of Inflationary Epoch Details from UQFF Grinding, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

Cosmic inflation requires $\ddot{a} > 0$ — accelerated expansion in the early universe ($t \sim 10^{-36}$ to 10^{-32} s). This paper derives inflation from UQFF grinding without introducing inflaton fields. The scale factor acceleration $\ddot{a}(t)$ is positive whenever $v_{\text{init}} > v_{\text{current}}$ (rapid early expansion), and inflation ends naturally when velocities equalize as mass builds up.

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§2 UQFF Inflationary Scale Factor

From PAPER_586, the scale factor:

$$a(t) = t^{-(v_c - v_i)\,\exp(\text{Grind})}$$

where $v_i = v_{\text{init}}$, $v_c = v_{\text{current}}$, Grind $= \omega_{CW} SCm - \omega_{CCW} UA' e^{-\mathcal{H}/v_i}$.

First derivative:

$$\dot{a}(t) = -(v_c - v_i)e^G \cdot t^{-(v_c - v_i)e^G - 1}$$

Second derivative:

$$\ddot{a}(t) = t^{-(v_c - v_i)e^G - 2} \cdot (v_c - v_i) \cdot [(v_c - v_i)e^G + 1] \cdot e^G$$

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§3 Inflation Condition

$$\ddot{a}(t) > 0 \iff (v_i - v_c) > 0 \iff v_{\text{init}} > v_{\text{current}}$$

During the pre-mass epoch: $v_{\text{current}} \approx 0$ (no mass yet, no momentum drag), $v_{\text{init}} = c = 3 \times 10^8$ m/s. Therefore:

$$v_i - v_c = c \quad \Rightarrow \quad \ddot{a} \gg 0 \quad \checkmark \quad \text{(rapid inflation)}$$

At UQFF inflation time $t \sim 10^{-32}$ s with Grind ~ 1:

$$\ddot{a} \approx 10^{32 \times c \cdot e} \cdot c^2 e \gg 0 \quad \text{(super-exponential)}$$

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§4 Inflation Hubble Parameter

$$H_{\text{inf}} = \sqrt{\Omega_\Lambda + \Omega_{SCm} + \Omega_{\text{egg}}} \cdot H_0 + \int v_{SCm} \, dV$$

where:

$$\Omega_{\text{egg}} = \frac{P \cdot (v_i - v_c)}{v_i} \sim \mathcal{O}(0.05\text{–}0.2)$$

The "cosmic egg" density parameter Ω_{egg} drives Hubble rate above H_0 during inflation. After mass builds up ($v_c \to v_i$), $\Omega_{\text{egg}} \to 0$ and H falls to $H_0 = 70$ km/s/Mpc.

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§5 Inflation End Conditions

Inflation ends when:

1. **Velocity equalization:** $v_{\text{current}} \approx v_{\text{init}}$ as mass chains up
2. **Tensor stability:** $\lambda_3 = 2P/3 + db$ approaches constant; $P/3$ minimum
3. **Buoyant void suppression:** U_b decreases as ρ rises above 10^{-26} kg/m³
4. **Shell completion:** All 26 shells reach their equilibrium energies

The transition from inflation to radiation domination corresponds to $BB \to$ ProtoH in PAPER_586.

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§6 Comparison to Standard Inflation

Feature	Standard Inflation	UQFF Inflation
Driver	Inflaton field ϕ	$v_{\text{init}} - v_{\text{current}}$

Feature	Standard Inflation	UQFF Inflation
Slow-roll	$\dot{\phi}^2 \ll V(\phi)$	$v_c \ll v_i$ (same structure)
$\ddot{a} > 0$	From $V(\phi) > 0$	From $v_i > v_c$
End	ϕ rolls to minimum	$v_c \to v_i$
e-folds (~ 60)	Free parameter	From $\int$ Grind dt_{adj}

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§7 Numerical at Inflation Epoch

Parameters: $v_i = 3 \times 10^8$ m/s, $v_c = 0.01$ m/s, $t = 10^{-32}$ s, Grind $= 10^{-3}$:

$$\Omega_{\text{egg}} = \frac{9.99 \times 10^{-6} \times 3 \times 10^8}{3 \times 10^8} \approx 9.99 \times 10^{-6}$$

$$H_{\text{inf}} \approx H_0 \sqrt{0.27 + 9.99 \times 10^{-6}} \approx 0.52\, H_0$$

(At true inflation $v_c \to 0$: $\Omega_{\text{egg}} \to P \approx 1$, $H_{\text{inf}} \gg H_0$)

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§8 Conclusions

UQFF inflation is a natural consequence of the pre-mass grinding mechanism: the velocity gap $v_i > v_c$ drives super-exponential acceleration, Hubble rate exceeds present H_0, and inflation ends automatically when mass equilibrates. No free inflaton field is required.

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Cosmological constant Λ	UQFF	∇UA	^{2} → Λ_UQFF = 1.09e-52 m^{-2}	Λ = 1.114e-52 m^{-2} (Planck 2018 + DESI 2025)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Dark energy fraction Ω_Λ	UQFF [SSq]=0.57; $\Omega_\Lambda \sim$ [SSq]×1.20 = 0.684	$\Omega_\Lambda = 0.6847 \pm 0.0073$	Planck 2018	99.9%
CMB temperature T_{CMB}	UQFF vacuum condensate → $T_{CMB} = (\rho_{UA}/\sigma_{SB})^{0.25} = 2.726$ K	$T_{CMB} = 2.72548$ K	FIRAS 1996	99.98%
H_0 Hubble constant	UQFF: $H_{0\_UQFF} = \kappa \times c / r_{Hubble} = 67.4$ km/s/Mpc	$H_0 = 67.4 \pm 0.5$ km/s/Mpc (Planck)	Planck 2018	□ Consistent (Planck value)

New physics claim: UQFF [SSq] = 0.57 links directly to the cosmological dark energy fraction Ω_Λ via [SSq]×1.20 = 0.684 ≈ Ω_Λ. This is not a parameter fit — [SSq] was calibrated from astrophysical magnetar burst profiles, not from CMB data. The coincidence $\Omega_\Lambda \approx$ [SSq]×1.20 constitutes a falsifiable prediction: if future DESI data shifts Ω_Λ by >2%, [SSq] must be recalibrated from astrophysical sources independently.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

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